

CSC10006 – Introduction to Database

Advanced Query

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1 Objectives

After completing this lab, students will be able to:

- Extended INSERT, UPDATE statements
- Use outer joins
- Query with CASE structure
- Use GROUP BY ... WITH CUBE | ROLLUP
- Use GROUPING SET and GROUPING function
- Use CASE structure with GROUPING

2 Detailed Instructions

2.1 Extended INSERT, UPDATE statements

Consider the following 2 data tables:

SINHVIEN (MASV, HOTEN, DIEMTB, HANG)

SINHVIENGIOI (MASV, HOTEN, DIEMTB)

2.1.1 UPDATE data from existing data:

Example 1: Update student ranking

```
update SINHVIEN
set HANG = (SELECT count(*)
           FROM SINHVIEN sv
           WHERE sv.DIEMTB >= SINHVIEN.DIEMTB)
```

2.1.2 INSERT data into a table from an existing table

Example 2: Add data to SINHVIENGIOI table for students with average score of 8.0 or higher

```

insert into SINHVIENGIOI
select MASV, HOTEN, DIEMTB
from SINHVIEN
where DIEMTB > 8

```

2.2 Joining

Consider the following 2 data tables:

SINHVIEN

	malop	tenlop
1	L1	10A
2	L2	10B
3	L3	10C

LOP

	masv	hoten	malop
1	01	A	L1
2	02	B	L2
3	03	C	L2
4	04	D	L1
5	05	E	L1

Requirement: Show the number of students in each class

2.2.1 Inner joins

Inner join between 2 tables A and B is a table $C = \{\text{tuples where each tuple is a combination of tuples in A with tuples in B such that the join condition is satisfied}\}$

To Inner join between SINHVIEN and LOP :

```

-- Using JOIN:
SELECT *
FROM SINHVIEN AS SV JOIN LOP AS L ON SV.MALOP = L.MALOP

-- Or using join condition in WHERE clause:
SELECT *
FROM SINHVIEN SV, LOP L
WHERE SV.MALOP = L.MALOP

```

Result :

	masv	hoten	malop	malop	tenlop
1	01	A	L1	L1	10A
2	02	B	L2	L2	10B
3	03	C	L2	L2	10B
4	04	D	L1	L1	10A
5	05	E	L1	L1	10A

Observation: Information about class 10C is lost

To Calculate number of student in each class :

```
SELECT L.MALOP, L.TENLOP, COUNT(*) AS SISO
FROM SINHVIEN AS SV JOIN LOP AS L ON SV.MALOP = L.MALOP
GROUP BY L.MALOP, L.TENLOP
```

Result :

	malop	tenlop	siso
1	L1	10A	3
2	L2	10B	2

Observation: The size of class 10C (equal to 0) is not displayed, because information about class 10C was lost after the inner join. Therefore, with inner join, tuples that do not satisfy the join condition will be removed. If you want it to appear → Use outer join.

2.2.2 Right (Outer) joins

Right Outer join between 2 tables A and B is a table C = {tuples where each tuple is a combination of tuples in A with tuples in B such that the join condition is satisfied} + {remaining tuples in B that do not satisfy the join condition with any tuple in A}

To Right (Outer) Joins between SINHVIEN and LOP :

```
SELECT *
FROM SINHVIEN as SV RIGHT JOIN LOP as L ON SV.MALOP = L.MALOP
```

Result:

	masv	hoten	malop	malop	tenlop
1	01	A	L1	L1	10A
2	04	D	L1	L1	10A
3	05	E	L1	L1	10A
4	02	B	L2	L2	10B
5	03	C	L2	L2	10B
6	NULL	NULL	NULL	L3	10C

Observation: Although class (10C) hasn't any student, but it still appear in result (because the Right Outer Join preserve the class 10C).

2.2.3 Left (Outer) joins

Left (Outer) join between 2 tables A and B is a table C = {tuples where each tuple is a combination of tuples in A with tuples in B such that the join condition is satisfied} + {remaining tuples in A that do not satisfy the join condition with any tuple in B}

2.2.4 Full (Outer) joins

Full Outer join between 2 tables A and B is a table C =

{tuples where each tuple is a combination of tuples in A with tuples in B such that the join condition is satisfied} +

{remaining tuples in A that do not satisfy the join condition with any tuple in B} +

{remaining tuples in B that do not satisfy the join condition with any tuple in A}

Consider the following 2 data tables:

PUBLISHER				
	pid	pname		pcity
1	1	Algodata Infosystems 1		MIA
2	2	Algodata Infosystems 2		NYO
3	3	Algodata Infosystems 3		MAN

AUTHORS

	auid	firstname	lastname	city
1	1	Reginald	Blotchet-Halls	NYO
2	2	Michel	DeFrance	OAS
3	3	Innes	del Castillo	CAN
4	4	Ann	Dull	LND
5	5	Marjorie	Green	CAL
6	6	Morningstar	Greene	CAL
7	7	Burt	Gringlesby	LOS
8	8	Sheryl	Hunter	NYO

SELECT *

FROM AUTHORS **FULL JOIN** PUBLISHERS ON city = pcityResult :

	auid	first...	lastname	city	pid	pname	pcity
1	1	Reginald	Blotch...	NYO...	2	Algoda...	NYO
2	2	Michel	DeFrance	OAS...	NULL	NULL	NULL
3	3	Innes	del Ca...	CAN...	NULL	NULL	NULL
4	4	Ann	Dull	LND...	NULL	NULL	NULL
5	5	Marjorie	Green	CAL...	NULL	NULL	NULL
6	6	Morni...	Greene	CAL...	NULL	NULL	NULL
7	7	Burt	Gringlesby	LOS...	NULL	NULL	NULL
8	8	Sheryl	Hunter	NYO...	2	Algoda...	NYO
9	NULL	NULL	NULL	NULL	3	Algoda...	MAN
10	NULL	NULL	NULL	NULL	1	Algoda...	MIA

2.3 Query with CASE structure

2.3.1 Structure 1

```
CASE input_expression
  WHEN when_expression THEN result_expression
  [ ...n ]
  [
    ELSE else_result_expression
  ]
END
```

Example 7: Show employee names and retirement year

```
SELECT HONV, TENNV,
(CASE PHAI
  WHEN 'Nam' THEN YEAR(NGSINH) + 60
  WHEN 'Nu' THEN YEAR(NGSINH) + 55
END ) AS NAMVEHUU
FROM NHANVIEN
```

Example 8: Show names of employees who have reached retirement age (male 60 years old, female 55 years old)

```
SELECT HONV, TENNV
FROM NHANVIEN
WHERE YEAR(GETDATE()) - YEAR(NGSINH) >= ( CASE PHAI
  WHEN 'Nam' THEN 60
  WHEN 'Nu' THEN 55
END )
```

2.3.2 Structure 2

```
CASE
  WHEN Boolean_expression THEN result_expression
  [ ...n ]
  [
    ELSE else_result_expression
  ]
END
```


Example 9: Show students and their academic performance classification

```
select MASV, CASE
    WHEN DIEMTB >= 8 THEN N'Giỏi'
    WHEN DIEMTB >= 7 THEN N'Khá'
    WHEN DIEMTB >= 6 THEN N'Trung bình khá'
    WHEN DIEMTB >= 5 THEN N'Trung bình'
END
from SINHVIEN
```

2.4 Use GROUP BY ... WITH CUBE | ROLLUP

2.4.1 GROUP BY ... WITH CUBE

Given the ITEM data table as follows:

	itemid	itemname	color	quantity
1	1	Table	Blue	124
2	2	Table	Red	223
3	3	Chair	Blue	101
4	4	Chair	Red	210

Example 10: Aggregate the quantity of items by name and color, by each name, by each color, total items.

```
SELECT ItemName, Color, SUM(Quantity) AS QtySum
FROM Inventory
GROUP BY ItemName, Color WITH CUBE
```

Kết quả là :

	ItemName	Color	QtySum
1	Chair	Blue	101
2	Chair	Red	210
3	Chair	NULL	311
4	Table	Blue	124
5	Table	Red	223
6	Table	NULL	347
7	NULL	NULL	658
8	NULL	Blue	225
9	NULL	Red	433

Số lượng Item tên Chair và màu Blue là 101

Số lượng Item tên Table (màu bất kỳ) là : 347

Số lượng Item (tên bất kỳ, màu bất kỳ) = tổng số item là 658

Số lượng Item màu Red (tên bất kỳ) : 433

Notes:

- Group by n attributes → will aggregate by 2^n criteria
- Aggregations with no data will not be displayed

2.4.2 GROUP BY ... WITH ROLLUP

Example 11: Aggregate the quantity of items by each name and color, by each name, total items

```
SELECT ItemName, Color, SUM(Quantity) AS QtySum
FROM Inventory
GROUP BY ItemName, Color WITH ROLLUP
```

Thống kê theo từng tên và màu, theo từng tên → tương tự **WITH CUBE** nhưng loại bỏ bớt tiêu chí thống kê : Các thống kê theo tên bất kỳ và một màu nào đó bị lược bỏ

Kết quả là :

	ItemName	Color	QtySum
1	Chair	Blue	101
2	Chair	Red	210
3	Chair	NULL	311
4	Table	Blue	124
5	Table	Red	223
6	Table	NULL	347
7	NULL	NULL	658

Số lượng Item tên Chair và màu Blue là 101

Số lượng Item tên Table (màu bất kỳ) là : 347

Số lượng Item (tên bất kỳ, màu bất kỳ) = tổng số item là 658

Lược bỏ 2 thống kê theo : (tên bất kỳ, màu đỏ) và (tên bất kỳ, màu xanh)

2.5 Use GROUP BY ... WITH CUBE | ROLLUP với GROUPING function

2.5.1 Case 1:

	itemid	itemname	color	quantity
1	1	Table	Blue	124
2	2	Table	Red	223
3	3	Chair	Blue	101
4	4	XYZ	Red	225

Example 12. Show the quantity of items for (each name, each color), (each name), (each color)

Kết quả :

	itemname	color	SoLuong
1	Chair	Blue	101
2	Chair	NULL	101
3	Table	Blue	124
4	Table	Red	223
5	Table	NULL	347
6	XYZ	Red	225
7	XYZ	NULL	225
8	NULL	NULL	673
9	NULL	Blue	225
10	NULL	Red	448

Số lượng item Chair (màu bất kỳ : NULL) là 101
→ Ở đây hiểu NULL là màu bất kỳ

NULL được hiểu là : Màu bất kỳ

Số lượng item (tên bất kỳ : NULL, màu bất kỳ : NULL) là 673

NULL được hiểu là : Màu bất kỳ

NULL được hiểu là : Tên bất kỳ

2.5.2 Case 2 (Modified data) → Problem ???

Modify data in the above table:

	itemid	itemname	color	quantity
1	1	Table	Blue	124
2	2	Table	Red	223
3	3	Chair	Blue	101
4	4	NULL	Red	225

Có một item có
itemname giá trị NULL

Result of re-executing the query:

	itemname	color	SoLuong
1	NULL	Red	225
2	NULL	NULL	225
3	Chair	Blue	101
4	Chair	NULL	101
5	Table	Blue	124
6	Table	Red	223
7	Table	NULL	347
8	NULL	NULL	673
9	NULL	Blue	225
10	NULL	Red	448

NULL được hiểu là tên bất kỳ
hay là itemname = NULL
→ ????

→ Cannot distinguish what NULL means here: is name equal to NULL or does it mean any name → Solution: **Use Grouping to determine**

2.5.3 Using GROUPING:

Syntax: **GROUPING (<attribute name>)**

Grouping is used to check whether the value of an attribute in the query result is from actual data or generated by the query

Explanation with the above example:

- If Grouping value of attribute name (itemname) is 1 → NULL value appears due to CUBE clause → meaning any name (generated by the query)
- If Grouping value of attribute name (itemname) is 0 → NULL value means the item's name is NULL (from actual data: itemname is NULL).

Example:

```
select itemname ,color, sum(quantity) as SoLuong,  
grouping (itemname) 'grp'  
from Inventory  
group by itemname, color with cube
```

Result:

	itemname	color	SoLuong	grp
1	NULL	Red	225	0
2	NULL	NULL	225	0
3	Chair	Blue	101	0
4	Chair	NULL	101	0
5	Table	Blue	124	0
6	Table	Red	223	0
7	Table	NULL	347	0
8	NULL	NULL	673	1
9	NULL	Blue	225	1
10	NULL	Red	448	1

NULL được hiểu là itemname = NULL
do Grouping(itemname)=0

→ Item NULL màu đỏ có số
lượng là 225

NULL được hiểu là tên bất kỳ do
Grouping(itemname) = 1

→ Item bất kỳ, màu đỏ có số
lượng là 448

2.6 Using Case ... End with Grouping

```
select (case
        when itemname IS NULL and
             grouping(itemname)=1 then N'Tên bất kỳ'
        else itemname
      end) as Itemname,
      (case
        when color is NULL and
             grouping(color)=1 then N'Màu bất kỳ'
        else color
      end) as Color,
      sum(quantity) as SoLuong
from Inventory
group by itemname, color with cube
```

Kết quả là:

	Itemname	Color	SoLuong
1	NULL	Red	225
2	NULL	Màu bất kỳ	225
3	Chair	Blue	101
4	Chair	Màu bất kỳ	101
5	Table	Blue	124
6	Table	Red	223
7	Table	Màu bất kỳ	347
8	Tên bất kỳ	Màu bất kỳ	673
9	Tên bất kỳ	Blue	225
10	Tên bất kỳ	Red	448

3 In-Class Exercises

Requirements: Write the following queries in SQL language, using the Project Management database.

- Q75. Show teacher names and department names they head if any
- Q76. Show list of department names and department head names if any
- Q77. Show list of teacher names and projects they lead if any
- Q78. Delete projects under topic "NCPT".

- Q79. Output teacher information (MAGV, HOTEN) and teacher's salary level. Salary level is classified by rule: Teacher salary < \$1800: "LOW"; From \$1800 to \$2200: AVERAGE; Salary > \$2200: "HIGH"
- Q80. Output teacher information (MAGV, HOTEN) and rank based on salary. If teacher has highest salary then rank is 1.
- Q81. Output teacher income information. Teacher income is calculated as SALARY + ALLOWANCE. If teacher is department head then ALLOWANCE is 300, and if teacher is faculty dean then ALLOWANCE is 600.
- Q82. Output the year the teacher is expected to retire with regulations: Retirement age for Male is 60, for Female is 55.

4 Homework Exercises

Requirements: Write the following queries in SQL language, using the Project Management database.

- Q83. Show the list of all teachers (magv, hoten) and the name of the teacher who is their academic supervisor.
- Q84. Show the list of all departments (mabm, tenbm), department head name and the number of teachers in each department.
- Q85. Show the list of all male teachers and information about the tasks they participated in.
- Q86. Show the list of all teachers and information about tasks in project 001 that they participated in.
- Q87. Show information about department heads (magv, hoten) who will retire in 2014. Note that the retirement age for male teachers is 60 and female teachers is 55.
- Q88. Show information about faculty deans (magv) and the year they will retire.
- Q89. Create table DANHSACHTHIDUA (magv, sodtbat, danhhieu) containing teacher ID, number of successful projects they participated in, and emulation title:

+ Insert data for this table (leave title column empty)

+ Based on sltdat column (number of projects with "pass" result) to update data for title column according to regulations:

- Sodttdat = 0 then title "not completed tasks"
 - $1 \leq \text{Sodttdat} \leq 2$ then title "completed tasks"
 - $3 \leq \text{Sodttdat} \leq 5$ then title "advanced"
 - Sodttdat ≥ 6 then title "excellent worker"
- Q90. Show magv, name and salary level of female teachers in "Information Technology" faculty, average salary, maximum and minimum salary of these teachers.
 - Q91. Show makhoa, tenkhoa, number of teachers in each faculty, average, maximum and minimum number of teachers in these faculties.
 - Q92. Show list of topic names, budget for topics (budget allocated to projects under the topic), total budget, maximum and minimum budget for topics.
 - Q93. Show madt, tendt, project budget, total and average budget of these projects by each project leader.
 - Q94. Show madt, tendt, project budget, total and average budget of these projects by each project level.
 - Q95. Aggregate the number of projects by (level, topic), by (level), by (topic).
 - Q96. Aggregate total salary of teachers by (department, gender), by (department).
 - Q97. Aggregate the number of teachers in IT faculty by (department, salary), by (department), by (salary).